Sustally

Scope 1 Cement Calculator

Comprehensive Guide to the User Interface and the Calculation Formulas

What every screen and field is, why it is there, and the exact formula behind every number.

Methodology pack: CSI Cement CO2 Protocol v2 | Scope 1 only | GWP AR5 / AR6

Contents

1. What this tool is

The Sustally Scope 1 Cement Calculator turns a cement plant's annual activity data (production, fuels, vehicles, refrigerants) into an audit-grade **Scope 1 greenhouse-gas inventory**. It is built on the CSI Cement CO2 Protocol with IPCC 2006 methodology, and it produces a downloadable report with a full calculation trace and the source of every factor.

It is a **Scope 1 only** tool. Scope 1 means direct emissions from sources the company owns or controls. Purchased electricity (Scope 2) and bought clinker / outsourced transport (Scope 3) are deliberately not collected here.

Who uses it: a sustainability lead or external consultant entering one facility's data for one reporting year, then exporting the report for disclosure or assurance.

2. Core principles you must understand

Four ideas explain almost every design choice in the UI:

2.1 The four Scope 1 categories

Every direct emission belongs to exactly one of four buckets. Step 4 is organised as four tabs for exactly this reason:

- **Process** - chemical CO2 from turning limestone into clinker (cement-specific, the largest source).
- **Stationary combustion** - burning fuels in the kiln and other fixed equipment.
- **Mobile combustion** - fuel burnt by owned/controlled vehicles and plant machinery.
- **Fugitive** - leakage of high-GWP gases (refrigerants, SF6 from switchgear).

2.2 Scope separation - never merged

Biomass CO2, combustion CH4/N2O, and anything outside Scope 1 are shown on separate lines and never folded into the headline number. This is a hard rule of GHG accounting and of the CSI protocol; the UI shows them in a distinct 'shown separately' panel.

2.3 Blank vs zero (this matters)

A blank field means **missing / unknown**. Typing **0** means a **confirmed actual zero**. They are treated very differently: a required value left blank can block or trigger a fallback, whereas 0 is a valid measured value. Every numeric field shows which state it is in.

2.4 Factor provenance & customisation

No factor is hidden. Every default carries its source, version and priority rank, and is recorded in the report's factor snapshot. A consultant can override any factor or fuel parameter, but must give a reason - which is also saved in the report.

3. Global screen elements

These appear on every step:

Element	What it is	Why it is there
Brand / title bar	Sustally logo and 'Scope 1 Cement Calculator'.	Identifies the product and the active sector pack.
GWP switch (AR5 / AR6)	Chooses which IPCC Global Warming Potential set converts CH_4, N_2O and refrigerants to CO_2e.	AR5 and AR6 give different CO_2e values; disclosure frameworks specify which to use, so it must be explicit and visible.
Theme toggle	Light / dark mode.	Comfort for long data-entry sessions; no effect on numbers.
Progress stepper	The 5 numbered steps; click to jump between them.	Shows where you are and lets you revisit earlier inputs without losing data.

4. Step 1 - Sector

You choose the industry methodology pack. **Cement** is active (CSI Cement CO_2 Protocol). Other sectors are shown but disabled.

Why it's there: the calculation core is sector-extensible. Sector selection loads the correct source categories, formulas and default factors before any data is entered, so the rest of the wizard is tailored to cement.

5. Step 2 - Organisation & boundary

Field	What it is	Why it is there
Company name	Legal entity preparing the inventory.	Printed on the report; identifies the reporting organisation.
Operating country	India or Other.	Drives region-appropriate factor defaults and a sanity warning if a non-local factor is used.
Consolidation / boundary method	Operational control, financial control, or equity share.	The GHG Protocol boundary rule that decides which sources count as the company's Scope 1. It must be declared before activity data is collected.

6. Step 3 - Facility & methodology selections

Facility name, type and reporting year identify the inventory. The methodology selections choose the calculation tier for each source. If the data a chosen tier needs is missing, the engine automatically steps down to the next-best method and records a warning - it never silently fails (see section 12).

Selection	Options	What it controls / why
Process method	CSI clinker-based; US EPA cement-based fallback	How process CO_2 is computed. CSI (clinker-based) is the accurate method; US EPA is a fallback when clinker tonnage is unavailable.
Clinker EF method	Plant-specific CaO/MgO; CSI default 0.525; IPCC default 0.510	The emission factor per tonne of clinker. Plant-specific is most accurate; the two defaults are fallbacks.
Dust method	Actual dust data; IPCC 2% fallback; Not applicable	How CO_2 from cement kiln dust (CKD) and bypass dust is handled.

Selection	Options	What it controls / why
Raw meal TOC method	CSI default TOC; Plant-specific TOC; Not applicable	CO_2 from organic carbon in the raw meal.
Fuel combustion method	Energy-based; Carbon-content-based; Direct measurement	How fuel CO_2 is derived. Energy-based (quantity x LHV x EF) is the default.

7. Step 4 - Activity data: the four categories

The activity step is four tabs. A counter on each tab shows how many entries it holds. Only the fields the selected methods actually need are shown.

7.1 Process tab

Production

Field	What it is / why
Clinker produced (t)	Tonnes of clinker made in the period. The basis of process emissions; required for the CSI method.
Cement produced (t)	Tonnes of finished cement. Used for the US EPA fallback and reference.
Cementitious product (t)	Total cementitious output. Used for the intensity metric (kgCO2e per tonne cementitious).

Clinker chemistry (shown only if Clinker EF method = Plant-specific)

Field	What it is / why
CaO %	Calcium oxide content of clinker. Drives the plant-specific emission factor.
Non-carbonate CaO %	Share of CaO from non-carbonate sources (e.g. slag) - subtracted so only carbonate-derived CO2 is counted.
MgO % / Non-carbonate MgO %	Same logic for magnesium oxide.

Dust (shown only if Dust method = Actual dust data)

Field	What it is / why
CKD leaving kiln (t)	Cement kiln dust not returned to the kiln; emits CO2.
CKD calcination rate (0-1)	Degree to which that dust is calcined. Default 1. Must be within 0-1.
Bypass dust leaving kiln (t)	Dust removed via the kiln bypass; emits CO2 at the clinker factor.
Bypass calcination rate (0-1)	Degree of calcination of bypass dust. Default 1.

Raw meal TOC (shown only if TOC method = Plant-specific)

Field	What it is / why
Raw meal / clinker ratio	Tonnes of raw meal per tonne of clinker. Default 1.55.
TOC fraction	Total organic carbon fraction of the raw meal. Default 0.002.

US EPA fallback (shown only if Process method = US EPA)

Field	What it is / why
Cement produced (t)	Used with the ratio to back-estimate clinker when clinker tonnage is unknown.
Clinker / cement ratio	Clinker fraction of cement.
Clinker EF override (tCO2/t)	Optional EF for the fallback; defaults to CSI 0.525.

7.2 Stationary combustion tab

Two lists: **Kiln fuels** and **Non-kiln fossil fuels**. Click 'Add fuel' to add a row. Each row:

Field	What it is / why
Label	Free text name for the entry (appears in the trace and report).

Field	What it is / why
Fuel	Picks the fuel; sets the default LHV, CO_2 EF and biomass fraction from the sourced library.
Category	Conventional fossil / Alternative fossil / Mixed / Biomass. Decides which Scope 1 bucket the CO_2 goes to and how the fossil/biomass split is applied.
Quantity	Amount of fuel burnt, in the fuel's unit (t, L, Sm3).
LHV override	Optional net calorific value (GJ/unit). Blank = use the sourced default.
CO2 EF override	Optional CO_2 emission factor (kg/GJ). Blank = use the sourced default.
Biomass frac	Optional 0-1 biogenic share for mixed fuels. Blank = library default.

Why kiln and non-kiln are separate: the report and trace distinguish conventional kiln fuel, alternative fossil kiln fuel and non-kiln fossil fuel, because assurance and benchmarking treat them differently.

7.3 Mobile combustion tab

Field	What it is / why
Label	Name of the vehicle / equipment group.
Ownership	Owned/controlled = counts in Scope 1. Third-party = excluded (kept out of the total).
Fuel	Fuel type; sets default LHV and EF.
Fuel quantity	Volume/mass of fuel consumed by that equipment.

7.4 Fugitive tab

Direct release of high-GWP gases - genuine Scope 1, reported as CO_2e using the selected GWP set.

Field	What it is / why
Label	Equipment description (e.g. plant chillers, HV switchgear).
Gas	Refrigerant or SF_6; sets the GWP from the gas library (AR5 or AR6).
Quantity leaked (kg)	Mass of gas released or topped-up over the period.
GWP override	Optional GWP (e.g. a supplier blend value). Blank = library GWP.

7.5 Customise factors (override panel)

Below the tabs, every constant factor is listed with its default value and source. Enter an override value **and a reason** to replace it. The override, the reason and 'priority rank 6 = user estimate' are written into the report's factor snapshot, so the change is fully auditable.

7.6 Live validation panel

As you type, the panel lists current errors (red, must fix) and warnings (amber, informational - usually 'a default was used'). It updates continuously so problems are caught before you calculate.

8. Step 5 - Results & downloadable report

Element	What it is / why
Gross Scope 1 hero	The headline total in tCO_2e (process + stationary + mobile + fugitive). The single number for disclosure.
Component cards	The nine Scope 1 components, each shown separately so the make-up of the total is transparent.
Shown separately panel	Biomass CO_2 (memo) and combustion CH_4/N_2O - reported but NOT inside gross, per protocol.
Intensity cards	Gross CO_2e per tonne clinker and per tonne cementitious - the benchmarking metrics.
Validation block	Final list of errors and warnings with their codes.
Download buttons	PDF report, Excel workbook (with full trace), JSON result. 'Save draft' stores it in the database.

9. The formula library

Every number the calculator produces comes from one of the formulas below. tCO2 = tonnes CO2; EF = emission factor; LHV = net calorific value.

9.1 Process emissions

Plant-specific clinker emission factor (tCO2 / t clinker)

```
correctedCaO = (CaO% - nonCarbonateCaO%) / 100
correctedMgO = (MgO% - nonCarbonateMgO%) / 100
Clinker EF = correctedCaO x 0.785 + correctedMgO x 1.092
```

Why it's there: 0.785 and 1.092 are the stoichiometric tonnes of CO2 released per tonne of CaO and MgO. Non-carbonate oxide is removed so only CO2 from limestone is counted. A negative corrected value is impossible and blocks the calculation.

Default clinker emission factors

```
CSI default = 0.525 tCO2 / t clinker
IPCC default = 0.785 x 0.65 = 0.510 tCO2 / t clinker
```

Why it's there: Used when plant chemistry is not available. CSI 0.525 is the protocol default; IPCC 0.510 assumes 65% CaO.

Clinker calcination CO2

```
Clinker calcination CO2 = Clinker produced (t) x Clinker EF
```

Why it's there: The dominant cement Scope 1 source: CO2 chemically released when limestone becomes clinker.

Bypass dust CO2

```
Bypass dust CO2 = Bypass dust (t) x Clinker EF x Bypass calcination rate
```

Why it's there: Dust removed through the kiln bypass has already released its CO2.

Cement kiln dust (CKD) CO2

```
fraction = (EFcli / (1 + EFcli)) x CKD calcination rate
CKD EF = fraction / (1 - fraction)
CKD CO2 = CKD quantity (t) x CKD EF
```

Why it's there: The CSI protocol formula for CO2 from kiln dust that is not returned to the kiln. The calcination rate must be within 0-1.

Dust 2% fallback

```
Dust CO2 = Clinker calcination CO2 x 0.02
```

Why it's there: IPCC default correction (+2%) used when actual dust data is unavailable.

Raw meal TOC CO2

```
Raw meal TOC CO2 = Clinker produced x RawMeal/Clinker ratio
x TOC fraction x (44 / 12)
```

Why it's there: CO2 from organic carbon in the raw materials. 44/12 is the IPCC/CSI mass ratio of CO2 to carbon. Defaults: ratio 1.55, TOC 0.002.

9.2 Combustion (stationary & mobile)

Energy-based fuel CO2 (default method)

```
Energy (TJ) = Quantity x LHV / 1000
Fuel CO2 (t) = Energy (TJ) x EF (kgCO2/GJ)
```

Why it's there: The standard activity-data method. LHV and EF come from the sourced fuel library unless overridden; a missing LHV with no default blocks the entry.

Carbon-content & direct methods

```
Carbon-content: CO2 (t) = Quantity (t) x CarbonFraction x (44/12)
Direct: CO2 (t) = metered tonnes entered directly
```

Why it's there: Alternatives to energy-based when carbon content or a CEMS reading is available.

Fossil / biomass split

```
Biomass CO2 = Total CO2 x BiomassFraction (memo, excluded)
Fossil CO2 = Total CO2 x (1 - BiomassFraction) (Scope 1)
```

Why it's there: Biogenic CO2 is a memo item under the GHG Protocol and must not inflate Scope 1. Mixed fuels (e.g. tyres ~27% biomass) are split automatically.

Mobile combustion (3 ways to get the fuel quantity)

```
Fuel-based: fuelQty = entered quantity
Hours-based: fuelQty = operating hours x consumption rate/hr
Distance-based: fuelQty = distance (km) x fuel per km
then Fuel CO2 = energy-based formula above
```

Why it's there: Owned/controlled equipment counts in Scope 1; third-party is excluded.

9.3 Fugitive emissions

Fugitive CO2e

```
Fugitive CO2e (t) = Leaked gas (kg) x GWP / 1000
```

Why it's there: Refrigerant/SF6 leakage. GWP is taken from the gas library for the selected AR5/AR6 set, or a per-entry override.

9.4 Separated (not in gross Scope 1)

Non-CSI combustion CH4 / N2O

```
CH4/N2O CO2e (t) = (EnergyGJ x CH4ef x GWP_CH4
+ EnergyGJ x N2Oef x GWP_N2O) / 1000
```

Why it's there: The CSI process method is CO2-only. Combustion CH4/N2O is computed and shown on its own line, never merged into the CSI CO2 total.

9.5 Totals & intensity

Gross Scope 1 and intensity

```
Gross Scope 1 = clinker calcination + bypass dust + CKD
+ raw meal TOC + conventional kiln fuel
+ alternative fossil kiln fuel + non-kiln fossil
+ mobile combustion + fugitive

Intensity = Gross Scope 1 x 1000 / Clinker produced (kgCO2e/t)
= Gross Scope 1 x 1000 / Cementitious product
```

Why it's there: The headline total is the simple sum of the nine separated components - nothing hidden, the sum is asserted to equal the displayed total.

10. Default factors and their sources

10.1 Methodology constants

Factor	Value	Unit	Source
CSI default clinker EF	0.525	tCO2/t clinker	CSI Cement CO2 Protocol v2.0
IPCC default clinker EF	0.510	tCO2/t clinker	IPCC 2006 (0.785 x 0.65)
CO2 per CaO	0.785	tCO2/tCaO	Stoichiometry
CO2 per MgO	1.092	tCO2/tMgO	Stoichiometry
CO2 per C	3.6667	tCO2/tC	IPCC/CSI 44/12 convention
Raw meal / clinker ratio	1.55	t/t	CSI default
TOC fraction	0.002	fraction	CSI default
CKD calcination rate	1	fraction	CSI default
Dust fallback	0.02	of calcination CO2	IPCC 2006 default

10.2 Fuel defaults (LHV and CO2 EF)

Fuel	Category	LHV	CO2 EF (kg/GJ)	Biomass frac
Coal (bituminous)	Conv. fossil	25.8 GJ/t	94.6	0
Petroleum coke	Conv. fossil	32.5 GJ/t	97.5	0
Lignite	Conv. fossil	11.9 GJ/t	101	0
Natural gas	Conv. fossil	0.0373 GJ/Sm3	56.1	0
Diesel / gas oil	Conv. fossil	0.0358 GJ/L	74.1	0
Heavy fuel oil	Conv. fossil	40.4 GJ/t	77.4	0
Waste oil	Alt. fossil	40.2 GJ/t	73.3	0
Used tyres	Mixed	28 GJ/t	85	0.27
Waste plastics	Alt. fossil	30 GJ/t	75	0
Mixed industrial waste	Mixed	18 GJ/t	83	0.50
Solid biomass	Biomass	11.6 GJ/t	112	1.00

Fuel CO2 EFs and LHVs follow IPCC 2006 Vol 2 Table 1.4; cement alternative-fuel values follow the CSI Cement CO2 Protocol. Every value is overridable with a reason.

10.3 Fugitive gas GWPs (100-year)

Gas	AR5 GWP	AR6 GWP
R-22 (HCFC-22)	1760	1960
R-32 (HFC-32)	677	771
R-134a	1300	1530
R-404A	3943	4728
R-407C	1624	1908
R-410A	1924	2256

Gas	AR5 GWP	AR6 GWP
R-507A	3985	4727
R-23 (HFC-23)	12400	14600
SF6 (switchgear)	23500	24300

Combustion gas GWPs: AR5 CH_4 28 / N_2O 265; AR6 CH_4 27 / N_2O 273.

11. Validation catalogue

Errors (red) block the result until fixed. Warnings (amber) are informational - most just say a default was used and are expected.

11.1 Common blocking errors

Code	Meaning
missing_clinker_production_for_csi_method	CSI method chosen but clinker tonnage is blank (and no US EPA inputs).
missing_cement_production_for_us_epa_fallback	US EPA fallback chosen but cement tonnage / ratio missing.
negative_corrected_cao / _mgo	Non-carbonate oxide exceeds total oxide - physically impossible.
ckd_calcination_rate_outside_0_1	CKD calcination rate not within 0-1.
missing_lhv_for_energy_based_fuel	Energy-based fuel with no LHV and no library default.
missing_fuel_quantity / _emission_factor	A fuel or fugitive row is missing its quantity / factor.
source_exclusion_without_reason	A source was switched off without a recorded reason.
missing_sector / reporting_period / facility / boundary_method	A required setup field is blank.

11.2 Common warnings

Code	Meaning
default_clinker_ef_used	A default clinker EF was used (no plant chemistry).
dust_2_percent_fallback_used	The IPCC 2% dust estimate was applied.
default_lhv_used / default_fuel_ef_used	A fuel's library LHV / EF default was used.
alternative_fuel_split_unknown	A mixed fuel's biomass split was not given; library default used.
default_toc_used / default_ckd_calcination_rate_used	A raw-meal or CKD default was used.

12. Automatic fallback logic

A missing input degrades to the next-best method with a warning - the calculation never simply fails:

- Clinker EF: Plant-specific CaO/MgO -> CSI default 0.525 -> IPCC default 0.510.
- Process: CSI clinker-based -> US EPA cement-based, automatically, if clinker tonnage is missing but cement + ratio are present.
- Dust: Actual dust data -> IPCC 2% fallback if no dust figures are entered.
- Fuel LHV / EF: entered value -> sourced library default (warning) -> blocking error only if no default exists.

Every fallback is recorded in the result's data-quality flags and in the calculation trace, so an auditor can see exactly what was substituted and why.

13. The three report exports

Export	Contents / use
PDF report	Two-page formal inventory: hero total, full Scope 1 breakdown, separated buckets, intensity, factor snapshots, validation and trace. For disclosure / sharing.
Excel workbook	Sheets: Summary, Factor snapshots (provenance), Calculation trace (every step, inputs and output), Warnings & errors. For audit and re-checking the maths.

Export	Contents / use
JSON	The complete structured result model. For integration or archiving.
Save draft to database	Stores the input payload, result, trace and factor snapshots in Payload (visible at /admin) so the run can be reopened and re-audited.

14. Glossary

Term	Meaning
Scope 1	Direct emissions from sources the company owns or controls.
Clinker	The intermediate product of a cement kiln; the main source of process CO_2.
Calcination	Heating limestone ($CaCO_3$) so it releases CO_2 and becomes CaO.
CKD	Cement kiln dust - fine dust captured from kiln gases.
TOC	Total organic carbon in the raw meal.
LHV	Lower (net) heating value of a fuel, GJ per unit.
EF	Emission factor - emissions per unit of activity.
GWP	Global Warming Potential - converts a gas to CO_2-equivalent.
CO_2e	CO_2-equivalent: all gases expressed on a common CO_2 basis.
Biogenic / biomass CO_2	CO_2 from biological material; a memo item, excluded from gross Scope 1.
CSI	Cement Sustainability Initiative - the cement CO_2 protocol used here.
Memo item	A figure reported alongside but deliberately outside the main total.

Sustally Scope 1 Cement Calculator - methodology-driven, audit-ready, Scope 1 only.